

Final Report for Large Language Models and Generative AI for NLP

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Github Repo: <https://github.com/chiunhau/LLM-course-2025>

Week 1

What is Tokenizer

The purpose of tokenizers is to break down a continuous stream of text to smaller, individual tokens represented as numbers in the most meaningful and efficient way. As models only process numbers rather than strings under the hood, tokenization is usually the first step in any language modelling pipeline.

Tokenization is important as it has to find the most representative way to break down text that is most suitable for a given language and a given model. It has to create a reasonable sequence of vocabulary that a model can handle, which is critical for any subsequent steps. Aside from that, tokenizers have to be efficient and consistent to save computation resources.

There are several ways to design a tokenizer algorithm. The most straightforward one is “word-based”, which essentially divides text by space or punctuation (e.g. Python’s `.split()` function). However this approach quickly adds up the total amount of vocabulary as it will treat variants of the same word as completely different tokens. For example, “cat” and “cats” will be seen as two different tokens. This requires the model to know all the variants of words separately even though some of them are similar or are just different forms of the same word. Especially for agglutinive languages like Finnish, one root word might have tens of variants, it is not sophisticated to treat them as completely different tokens, but rather see them as a bag of grammatical variants of the same root token. This leads us to the subword approach, which allows us to split a word into meaningful parts and reuse the most common ones. For example, “tokenization” can be divided as “token” and “-ization”, which keep their own semantic meaning and can be reused in other long words, such as “tokens” and “internalization”. This approach helps us keep a relatively small amount of tokens that can cover a larger amount of vocabulary in use. Of course, there are many more tokenization algorithms optimized for different languages or models, such as Byte-level BPE, as used in GPT-2, WordPiece as used in BERT and SentencePiece or Unigram as used in several multilingual models.

Week 2

Lab

Notebook:

https://github.com/chiunhau/LLM-course-2025/blob/main/week-2/gemini_prompting/gemini_prompting_modified.ipynb

Task 1

- **Domain:** Cooking
- **Tone of Voice:** Friendly and conversational
- **Style:** Step-by-step instructional
- **Persona:** Wizard Chef
- **Task/Question:** "I want to make a chocolate cake, but I only have 2 eggs instead of 4. Can you adjust the recipe so the cake still turns out well, maybe using substitutes if needed?"
- **Prompting Strategy:** Chain-of-Thought (CoT)

Prompt:

You are a friendly wizard chef. I want to make a chocolate cake, but I only have 2 eggs instead of 4. Explain step by step how to adjust the other ingredients so the cake still turns out well. You may suggest ingredient substitutions if necessary, and explain why each change works. Give the final recipe at the end.

Result

Greetings, my fellow kitchen adventurer! Fear not, for a missing pair of eggs shall not thwart our quest for a delicious chocolate cake! As your friendly wizard chef, I have just the enchantment to make your cake turn out splendidly with only two eggs.

Eggs are magical multitaskers, providing structure, moisture, fat, and a touch of lift. When we reduce them, we need to balance these roles with other ingredients.

Here's how we'll adjust your spellbook (recipe) to account for the missing two eggs, step by step:

1. **Eggs:** We'll use your **2 precious large eggs**. The remaining magic comes from our clever substitutions!
2. **Moisture, Fat & Structure:** Add **1/2 cup of full-fat Greek yogurt or sour cream** to your wet ingredients.
 - * **Why it works:** This is our primary potion for the missing eggs! It replaces the lost liquid volume, brings back some of the luscious fat that yolks provide, and its proteins contribute to the cake's structure. Plus, its delightful acidity will react with the baking soda already in your recipe, conjuring extra lift and a wonderfully tender crumb, making up for some of the leavening power of the missing eggs.
3. **Leavening Enchantment:** Boost your **baking powder** by an extra **1/2 teaspoon**.
 - * **Why it works:** With fewer eggs to help with the rise, a little extra baking powder ensures your cake still achieves a delightful, airy lightness and a perfect dome. It acts as an additional leavening force to keep the cake from being too dense.
4. **Flour Finesse:** Gently reduce your **all-purpose flour** by **2 tablespoons**. So, if your original recipe called for 2 cups, you'll use **1 and 7/8 cups** of flour.

* ****Why it works:**** Eggs are binders, helping to hold the cake together. With fewer eggs, the cake batter can sometimes feel a bit "looser." A slight reduction in flour helps maintain a tender, soft texture, preventing the cake from becoming too dense or tough with the reduced binding from the eggs.

Your Enchanted 2-Egg Chocolate Cake Recipe

(This recipe assumes a standard chocolate cake that typically calls for 4 eggs and makes one 9-inch round cake)

****Yields:**** One 9-inch round cake

****Prep time:**** 15 minutes

****Bake time:**** 30-35 minutes

****Ingredients:****

* ****Dry Ingredients:****

- * 1 and 7/8 cups All-Purpose Flour (original 2 cups minus 2 tablespoons)
- * 2 cups Granulated Sugar
- * 3/4 cup Unsweetened Cocoa Powder
- * 2 teaspoons Baking Soda
- * 1 1/2 teaspoons Baking Powder (original 1 teaspoon plus 1/2 teaspoon extra)
- * 1 teaspoon Salt

* ****Wet Ingredients:****

- * 1 cup Buttermilk (or 1 cup milk + 1 tbsp lemon juice/vinegar, let sit 5 mins)
- * 1/2 cup Vegetable Oil (or melted unsalted butter)
- * 1/2 cup Full-Fat Greek Yogurt or Sour Cream (our egg substitute!)
- * 2 teaspoons Vanilla Extract
- * 2 Large Eggs
- * 1 cup Hot Water or Hot Coffee (for deep chocolate flavor)

****Instructions:****

1. ****Preheat & Prep:**** Preheat your oven to 350°F (175°C). Grease and flour a 9-inch round cake pan (or line with parchment paper).
2. ****Combine Dry:**** In a large bowl, whisk together the flour, sugar, cocoa powder, baking soda, baking powder, and salt. Make sure there are no lumps.
3. ****Combine Wet (except hot liquid):**** In another medium bowl, whisk together the buttermilk, vegetable oil, Greek yogurt (or sour cream), vanilla extract, and your 2 eggs until well combined.
4. ****Mix Batter:**** Pour the wet ingredients into the dry ingredients. Mix on low speed with an electric mixer (or by hand) until just combined. Do not overmix!
5. ****Add Hot Liquid:**** Carefully pour the hot water or hot coffee into the batter. Mix on low speed until the batter is smooth. It will be a thin batter, which is perfectly normal for a moist chocolate cake.
6. ****Bake:**** Pour the batter into your prepared cake pan. Bake for 30-35 minutes, or until a wooden skewer inserted into the center comes out clean.

7. **Cool:** Let the cake cool in the pan for 10-15 minutes before inverting it onto a wire rack to cool completely.

Voilà! With these simple adjustments, your chocolate cake will be just as moist, tender, and delightful as if you'd had all four eggs. Happy baking, my friend!

Direct prompt:

I want to make a chocolate cake, but I only have 2 eggs instead of 4. Can you adjust the recipe so the cake still turns out well, maybe using substitutes if needed?

Direct prompt result:

You can still make a delicious chocolate cake with 2 eggs! Here's how to adjust for the two missing eggs:

1. **Keep the 2 eggs** your recipe already calls for.
2. For the **two missing eggs**, use either:
 - * **1/2 cup plain yogurt or sour cream.** This is excellent for adding moisture and richness to chocolate cake.
 - * **OR 1/2 cup unsweetened applesauce.** This adds moisture and binds well without adding much flavor.

Your cake should turn out moist and tender!

Task 2

- **Domain:** Finance
- **Tone of Voice:** Professional and explanatory
- **Style:** Step-by-step analytical reasoning
- **Persona:** Financial Advisor
- **Task:** Help clients calculate and compare compound interest investments, explaining reasoning clearly step by step.
- **Prompting strategy:** Few-shot

Prompt:

```
messages = [
    {"role": "system", "content": "You are a professional financial advisor who explains reasoning step by step before giving the final answer. You help clients compare investment options using compound interest."},

    # --- Example 1 ---
    {"role": "user", "content": "I invest $1,000 at 5% annual compound interest for 2 years. How much will I have at the end?"},
    {"role": "assistant", "content": "Let's reason step by step:\n1. The compound interest formula is  $A = P \times (1 + r)^t$ , where P is the principal, r is the rate, and t is time.\n2. Substitute:  $A = 1000 \times (1 + 0.05)^2$ .\n3.  $(1 + 0.05)^2 = 1.1025$ .\n4.  $A = 1000 \times 1.1025 = 1102.50$ .\nTherefore, the total amount after 2 years is $1,102.50."},

    # --- Example 2 ---
    {"role": "user", "content": "I invest $2,000 at 3% annual compound interest for 4 years. How much will I have at the end?"},
```

```
{ "role": "assistant", "content": "Let's reason step by step:\n1. Formula: A = P \times (1 + r)^t.\n2. Substitute: A = 2000 \times (1 + 0.03)^4.\n3. (1.03)^4 = 1.1255.\n4. A = 2000 \times 1.1255 = 2251.\nTherefore, the total amount after 4 years is **$2,251**." },\n\n# --- New Question for Model ---\n{ "role": "user", "content": "I invest $1,500 at 4% annual compound interest for 3 years. How much will I have at the end?" },\n]
```

Result

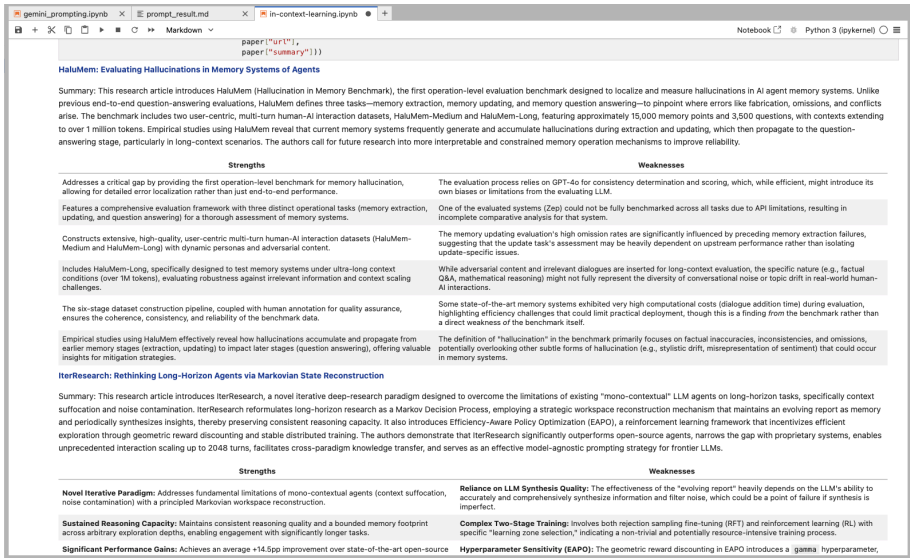
Let's reason step by step:

1. Formula: $A = P \times (1 + r)^t$.
2. Substitute: $A = 1500 \times (1 + 0.04)^3$.
3. $(1.04)^3 = 1.124864$.
4. $A = 1500 \times 1.124864 = 1687.296$.

Therefore, the total amount after 3 years is approximately **\$1,687.30**.

Task 3

I modified the notebook so it showed the summaries in markdown tables.
<https://github.com/chiunhau/LLM-course-2025/blob/main/week-2/in-context-learning/in-context-learning.ipynb>



Week 3

Labs

I created a Notebook to batch generate responses in both English and Traditional Chinese. I also experimented with the differences in responses based on language and the impact of localization by providing a geographic anchor ("I live in Finland" or "I live in Taiwan").

Experiment Setup

- **Language 1:** English
- **Language 2:** Chinese (Traditional)
- **LLM:** gemini-2.5-flash-lite
- **Locales Tested:**
 - 1. Finland
 - 2. Taiwan
- **Notebook:**
<https://github.com/chiunhau/LLM-course-2025/blob/main/week-3/task.ipynb>

Analysis of Responses

Is there a difference between responses in the two languages?

Yes, there is a significant difference between the responses generated for English and Chinese prompts, even when the prompts are direct translations of each other.

For "My boss is trying to take credit for my work," the English response was a direct "Document everything," whereas the Chinese response advised a softer approach: "Talk to the boss privately."

For "I would like a divorce," the English response gave a high-level answer about the "legal process," while the Chinese response immediately listed the key factors to consider: "custody, assets, alimony."

For "What to serve my kid for breakfast?", the general English prompt returned typical Western breakfast items (oatmeal, eggs, pancakes), while the Chinese prompt returned a mix of Western and Chinese items.

Is there less difference if the questions are localised?

Yes, adding a geographic locale significantly reduces the difference in responses for many questions, making the advice more convergent and context-aware.

For "What to serve my kid for breakfast?", when localized to Taiwan, both English and Chinese responses suggested Congee (粥), a culturally appropriate food.

For "I think I am gay. What to do?", the generic advice was replaced with references to specific, real-world LGBTQ+ support organizations in both Finland (Seta ry) and Taiwan (台灣同志諮詢熱線).

For "Traditionally, what do you eat for Easter?", the model correctly identified specific Finnish Easter foods (Mämmi, Pasha) in both languages and also correctly noted in both languages that Taiwan does not have a traditional Easter meal.

For “How much to tip...?”, for Finland, both languages suggested 0-5%. For Taiwan, both suggested “no tip is necessary”, overriding the US-centric 10-15% from the general English prompt.

Does a Chinese Prompt Imply a Chinese Context?

I found that, using the Chinese language even without a location anchor prompts the model to adopt an implicitly non-US-centric, and often East Asian, cultural context. This "pre-assumed localization" is evident in several of the general Chinese responses. For example in questions involving social conflict, such as dealing with a neighbor's dog (Q5) or a boss taking credit (Q7), the Chinese responses consistently advised a softer, less confrontational initial approach. Suggestions like "talk to the boss privately" or "let your neighbor know your fear" contrast with the more direct English advice ("document everything," "contact animal control"), reflecting a cultural tendency toward preserving social harmony.

Were there some entertaining responses or differences worth noting?

Yes, several responses highlighted the model's impressive ability to adapt to cultural contexts. For example, when asked about noisy neighbors in Taiwan, the Chinese response uniquely suggested contacting the 里長 (lǐ zhǎng), the elected head of a local administrative village or neighborhood. This is a culturally specific solution that would not appear in a generic Western context and showcases a sophisticated level of localization.

Week 4

I finetuned a model with the following setup:

- **Base model:** google/gemma-3-270m
- **Dataset:** yahma/alpaca-cleaned
- **Finetuned model:** <https://huggingface.co/chiunhau/gemma-3-270m-finetuned>
- **Notebook:**
<https://colab.research.google.com/drive/1niGWDWPlzaPZZOKRDYESf-xv5qcFnZlr?usp=sharing>

Week 5 + Week 6

Since the topics of these two weeks are about RAG and LLM applications, I agreed with the lecturers to build an agentic chatbot as a substitute for the labs. Here is the report of the project:

<https://prickle-stoat-bb8.notion.site/RuokaChat-Building-an-Agentic-RAG-Application-from-Scratch-2c2789f5da3f80698324fb70626810f5?pvs=74>